

Literature status for arXiv:0809.2723, Question 7.5

Run `fa_banach_001`

2026-06-14

Status

This is a lightweight `literature_already_answered` packet. It records that Question 7.5 from Avilés, Kadets, Martín, Merí, and Shepelska, *Slicely Countably Determined Banach spaces*, arXiv:0809.2723, was explicitly answered in later literature by Kadets, Pérez, and Werner, *Operations with slicely countably determined sets*, arXiv:1708.05218.

Source question

Question 7.5 of arXiv:0809.2723 appears on page 27 of the source PDF and at TeX source lines 2425–2428:

Are the concepts of SCD sets and almost-SCD sets equivalent?

The question refers to Remark 4.5 of the same source paper for the definition of almost-SCD.

Supporting answer

The supporting paper explicitly cites the source question as [2, Question 7.5] and states in its introduction that the properties aSCD and SCD are not equivalent for general bounded closed convex sets. The negative answer is given by [2, Theorem 4.3]: if X is a Banach space with the Daugavet property, then there exists a convex closed bounded set $\tilde{C} \subset X$ which is aSCD but not SCD.

The same paper also proves the balanced-set refinement [2, Theorem 4.5]: every convex closed bounded balanced aSCD set is SCD. Thus Question 7.5 is answered negatively in general, while the balanced case relevant to many unit-ball applications has a positive answer.

Scope and novelty

This packet records an exact later-literature answer, not a new proof from the run. Local duplicate checks over the run registry, solution, attempt, and proof-gap indexes found no existing packet for arXiv:0809.2723 / arXiv:1708.05218 or the almost-SCD/aSCD question.

References

- [1] A. Avilés, V. Kadets, M. Martín, J. Merí, and V. Shepelska, *Slicely Countably Determined Banach spaces*, arXiv:0809.2723.

- [2] V. Kadets, A. Pérez, and D. Werner, *Operations with slicely countably determined sets*, arXiv:1708.05218.